Annotating the conceptual domain of food across different contexts

These annotation guidelines are intended to annotate the conceptual domain of food across contexts from a historical point of view. The guidelines are built upon FrameNet.

FrameNet Recap

LUs and core and peripheral FEs

These guidelines are inspired by FrameNet annotation, particularly the FrameNet Brasil extension. The core idea behind FrameNet is that events and situations are represented through Frames (schematic representations of scenarios) involving various participants, objects, and conceptual roles, each of which is a Frame Element (FE). A Frame is triggered by a lexical unit (LU) or a multiword expression (a lexical unit composed of more than one lexeme). A lexical unit is defined as the pairing of a lemma and a Frame, meaning a word considered in one of its specific senses, allowing polysemous words to be distinguished according to the Frame they evoke.

For example, understanding the meaning of *buy* in the sentence *'Daniel bought a new car'* requires a cognitive model of buying, selling, and commodity exchange, which is structured through specific roles (FEs) that help define the scenario. In this case, *bought* is the <u>Lexical Unit</u> that evokes the **Commerce_buy** Frame, with its corresponding FEs: *Buyer* (Daniel) and *Goods* (a new car).

(1) Daniel <u>bought</u> a new car

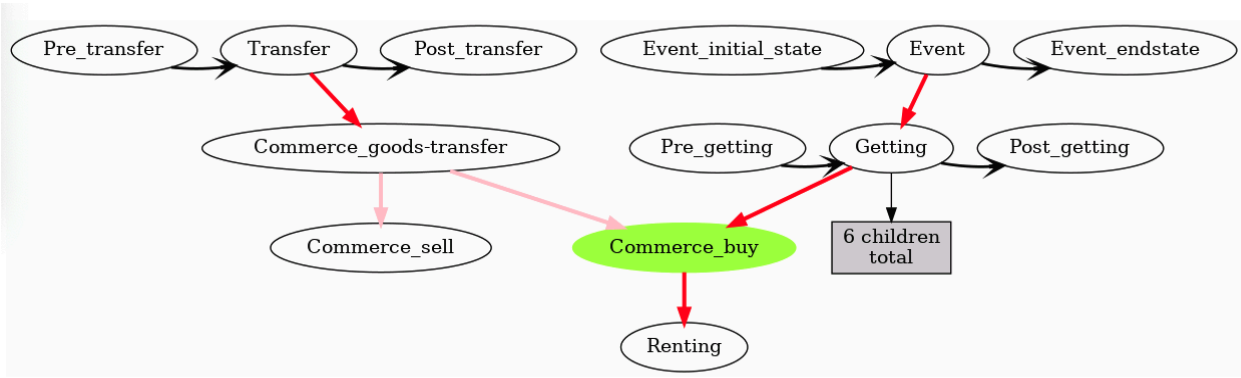
Buyer and *Goods* are defined as Core Frame Elements, meaning that they are essential to the meaning of a Frame. Together with the Core FEs, each Frame can also possess Non-Core (or peripheral) FEs. These Peripheral FEs provide additional contextual information about an event but are not essential to its basic structure. Unlike Core FEs, which define the fundamental participants in a Frame, non-core FEs typically express details such as time, place, manner, or reason.

For example, in the **Commerce_buy** Frame, *Buyer* and *Goods* are core FEs because they are necessary to define the act of buying. In contrast, an element like *Time* (e.g., *'I bought a car [yesterday]'*) or *Means* (e.g., *'I bought a car [with a loan]'*) is non-core, as it enriches the scenario but is not required to understand the event itself.

Relations in Berkeley FrameNet

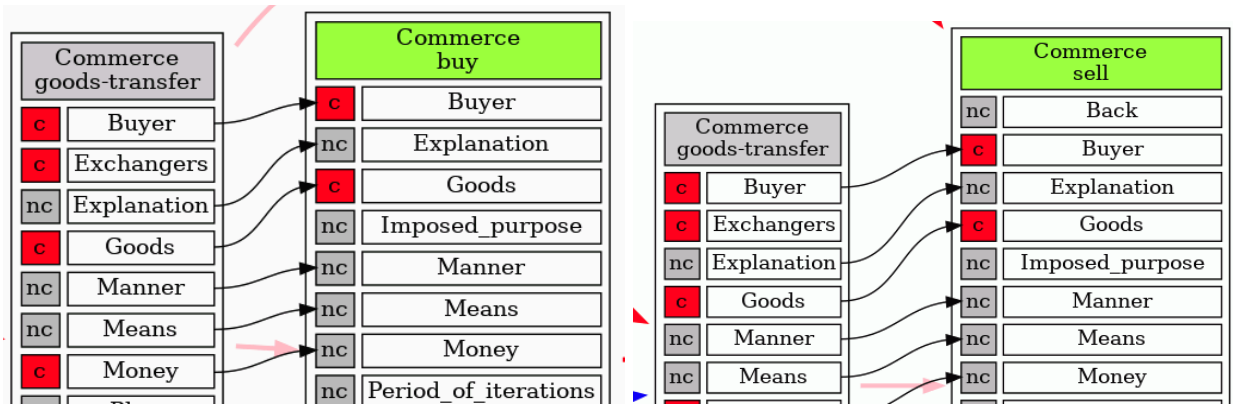
In FrameNet, Frames are also connected through specific relationships to capture how different conceptual structures are interconnected (both as Frame-to-Frame and FE-to-FE). These include hierarchical links like **inheritance** (where one Frame is a more specific version of another) and associative links such as **perspective on** (which highlight dependencies or alternative viewpoints on the same event). In Fig. 1, we provide the visualisation of some of these relations as represented in FrameNet.

FrameNet includes 8 possible relations: (1) *Inheritance* – one Frame is a more specific version of another, inheriting its structure and meaning (e.g., **Commerce_buy** inherits from **Commerce_goods_transfer**); (2) *Subframe* – a complex event is broken into smaller Frames representing its stages (e.g., **Criminal_process** includes **Arrest, Trial**, and **Sentencing** as subframes); (3) *Using* – A Frame relies on another for part of its meaning but does not inherit from it (e.g., **Legal_authority** uses **Social_agreement**); (4) *Perspective On* – Two Frames describe the same event from different viewpoints (e.g., **Commerce_buy** and **Commerce_sell** are perspectives on **Commerce_goods_transfer**); (5) *Inchoative Of* – One Frame represents the beginning of a state described by another (e.g., **Becoming_wet** is the inchoative of **Being_wet**); (6) *Causative Of* – One Frame describes an action that causes another Frame's state or event (e.g., **Cause_to_be_wet** is the causative of **Being_wet**) (7) *See Also* – Frames are linked due to conceptual similarity but without a strict structural relationship (e.g., **Attack** and **Defense**); (8) *Precedes* – One Frame represents an event that typically happens before another (e.g., **Court_trial** precedes **Verdict**).



The relations between Frames operate at a general level (Frame-to-Frame), through a FE-to-FE (Frame Element to Frame Element) systematic mapping between related Frames. When two Frames are connected through relations like *Inheritance*, *Subframe*, *Perspective On*, *Causative Of*, or *Inchoative Of*, their Frame Elements often correspond in a structured way. For example, **Commerce_goods_transfer** Frame maps its FEs through an inheritance relation on both the **Commerce_buy** Frame and the ***Commerce_sell***

Frame, as illustrated in Fig. 2.



These relations are able to capture the great majority of relations but fall short in three cases: in FE-to-Frame relations, in metonymic realizations of frames, and in qualia relations. These problems were addressed in the construction of FrameNet Brasil.

Relations in FrameNet Brasil

In FrameNet Brasil (FN-Br), the authors implemented qualia roles, integrating the qualia directly into the FN-Br database (Torrent et al., 2023). Based on the Generative Lexicon (GL) Theory by Pustejovsky (1991, 1995), this approach treats lexical items as complex information structures defined across four levels: Argument structure (A), Event structure (E), Qualia structure (Q), and Lexical Inheritance structure (I). We are interested in this study to the qualia structure which encodes semantic dimensions essential to meaning, through four roles:

1. *Formal*: values establishing what differentiates a given object within its semantic domain; it is typically the description of its basic category.

2. *Constitutive*: values expressing the relationship between a given object and its constituents or parts, such as material, weight or characteristic parts.

3. *Telic*: values related to the function or purpose of an object, such as the intention of an agent performing a given action or the intrinsic function of an object.

4. *Agentive*: values determining the origin of the object, such as its creator, type of origin (natural or artificial) or its initial cause.

In Torrent et al., (2022), the authors presented a visualisation adapted by Belcavello et al. (2020) of the word *pizza* through qualia roles. We report the picture below (Fig. 1).

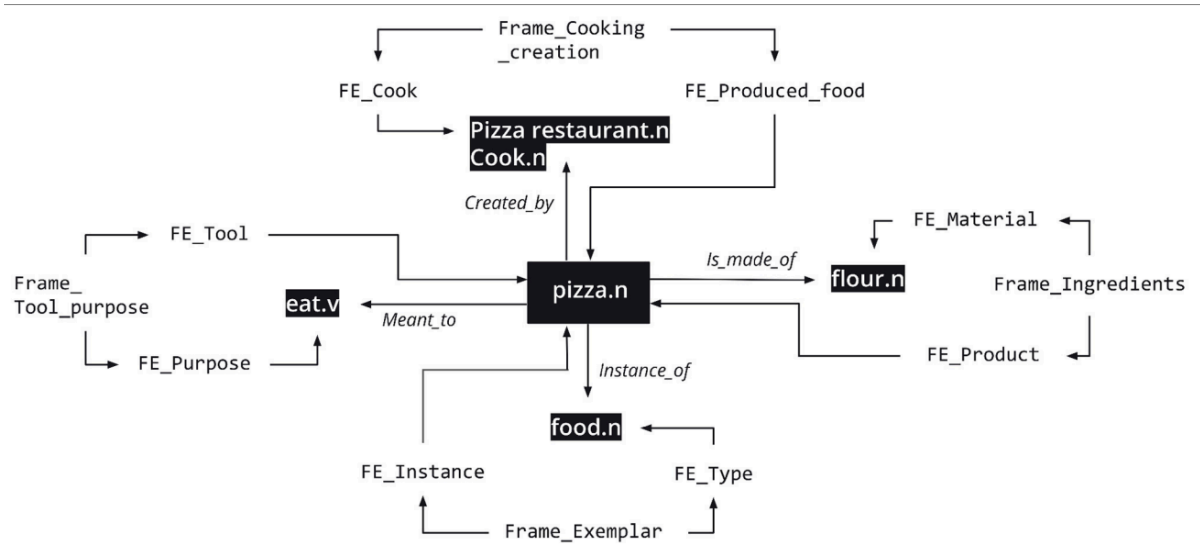
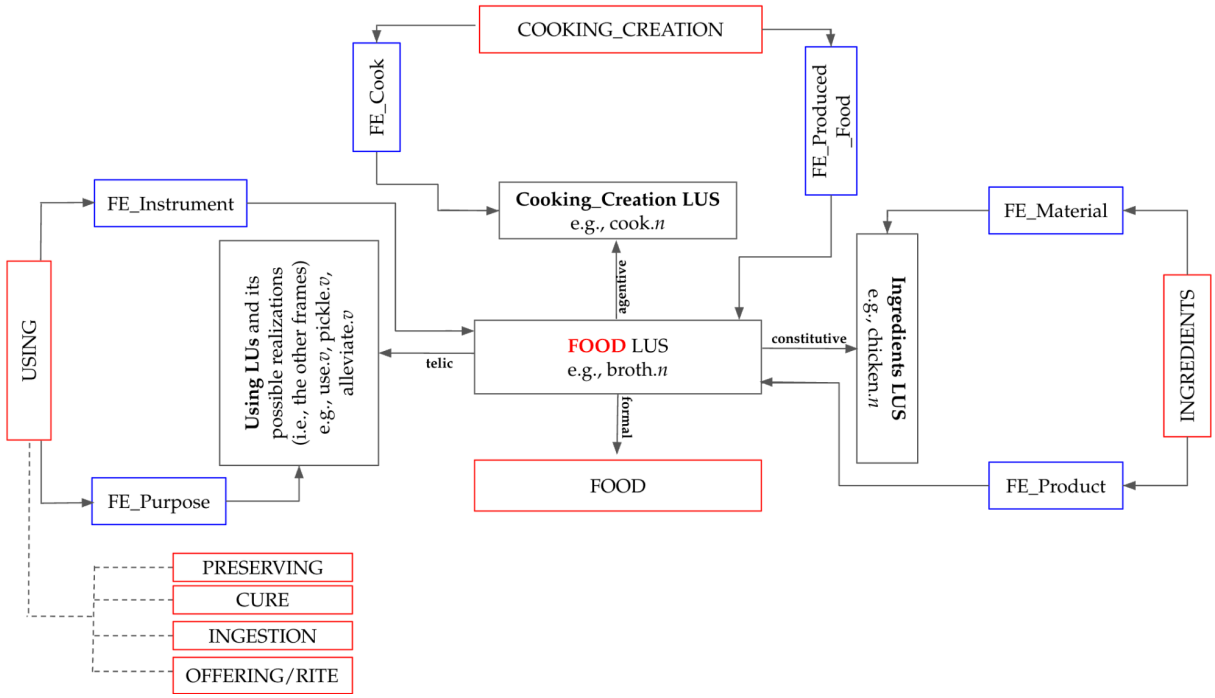


Fig. 1

Food in context

We are interested in capturing food from an historical perspective as well as considering any conceptualization of food across different contexts. We do so, through an adaptation of the qualia relations proposed in Torrent et al. (2022, 2023) and Belcavello et al. (2020). The purpose was to represent food in its realizations with a focus on its uses and purposes different from nutrition. We express this through an adaptation of the ternary qualia relations, as illustrated in Fig. 2. In the following paragraph, we present the characterization of each frame through the specific qualia relations and their annotation process.



Annotation

The annotation of food across different contexts is conducted with INCEpTION (Klie at al., 2019). The first step of the annotation is the identification of the Lexical Unit of **Food**, namely every word referring to a food entity. An initial list of Lexical Units is provided below, with a particular attention for historical varieties, such as historical spelling, or obsolete words.

List

The annotator has to select the word and choose the correct label Food_LU which stands for Lexical Unit Food. The annotator has to preferably annotate just the word referring to the food without the article, unless the article or any other element is necessary to define the entity.

Once you identified the LU, the possible frames which are activated are:

FOOD: the exemplar frame

Food frame is activated by a LU of **Food** through the *formal* qualia relation. This Frame is defined in FrameNet as words referring to items of food, and it is considered the

exemplar frame in this schema, which means its LUs are the ones which we consider as starting point to activate all the possible relations with the other frames. E.g., the word 'broth' in the flowchart is a LU of the **Food** frame. This Frame in FrameNet has 4 Frame Elements, Food (core), Constituent_parts, Descriptor and Type (non-core). We decided that for the purpose of this study we do not need to distinguish between Descriptor and Type since they are very similar concepts and we are just interested in the general characterization of food. We then aggregate the two under Descriptor, and we add a second level of annotation for the intrinsic polarity of Descriptors (negative, positive, and neutral). Since we are interested in the characterization of food in different contexts and the habits related to its use, we introduced other two FEs to trace its quantity, a possible marker for alternative uses of food as shown in a previous study (Bhagwat et al., 2025). We took these FEs from **Measures** frame. The final frame in our schema is then structured as:

<u>Core</u>:
Food [FOOD_LU]: corresponds to the LUs
<u>Non-core</u>:

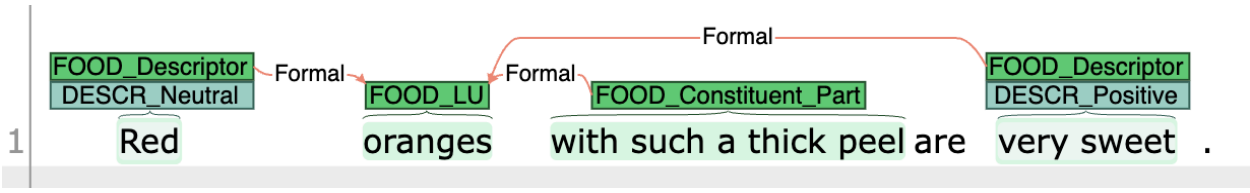
(1) *Constituent_parts* [FOOD_Constituent_Part]: a part of the *Food_LU*; (???)

(2) *Descriptor* [FOOD_Descriptor]: a characteristic or description, as well as a particular type of the *Food_LU*; which can be (1) *Positive* [DESCR_Positive]; (2) *Negative* [DESCR_Negative]; (3) *Neutral* [DESCR_Neutral].

(3) *Count* [FOOD_Count]: the number of units of the measured *Food_LU*;

(4) *Unit* [FOOD_Unit]: the standardized quantity of *Food_LU* in terms of which other quantities of the same kind are measured.

Once you have identified the LU of Food and its Frame Elements, by clicking on the word or the multiword expression, from the **Label** menu under the **Layer Span** you need to annotate the *formal* relation selected by the **Layer Relation** and link each of the FEs of **Food** to the FOOD_LU.



INGREDIENTS: the constitutive frame

Ingredients Frame is activated by any LU expressing any food material which is used to

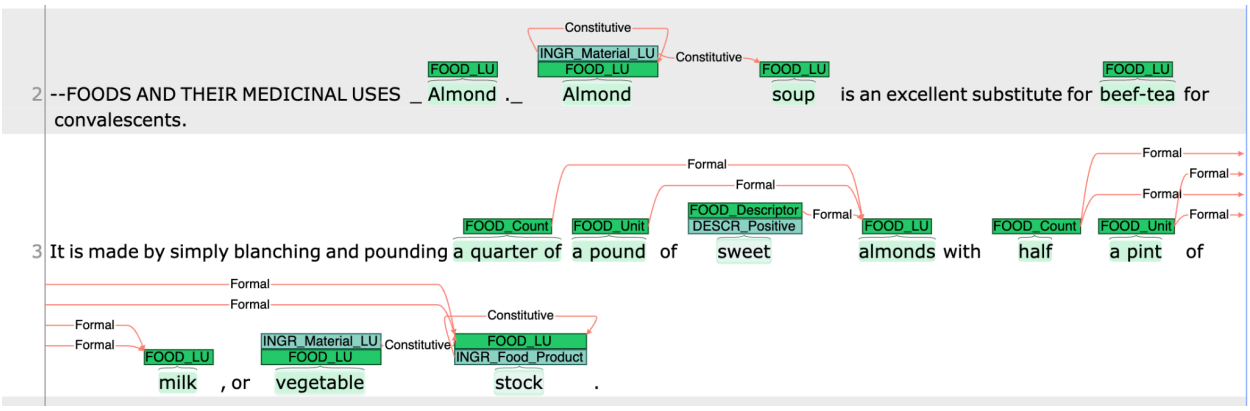
create a food product through the *constitutive* qualia relation. This Frame is present in FrameNet Brasil but not in the Berkeley one. This Frame is defined in FrameNet Brasil as a Material used to make a Product_Food, thus in this case any food which is used to create another. This Frame in FrameNet has 7 Frame Elements, Material, Product and Use (core), Descriptor, Name, Type and Origin (non-core). For the purpose of the present study we aggregate again Descriptor and Type and we excluded Origin. This frame has to be annotated only when it is clearly distinguished by Food, otherwise the Food annotation has the precedence.

<u>Core</u>:

(1) *Material* [INGR_Material_LU]: The entity which is used to make a Product_Food.

(2) *Product* [INGR_Food_Product]: The entity or substance that is created out of the Material, in this case the food.

(3) *Use* [INGR_Use]: The purpose for which the Material is intended.



Notice that the annotator has to double annotate the FOOD_LU both in its being the exemplar frame LU and the specific FE of the other activated frame (in this case Product_Food of **Ingredients**). In the list of labels in INCEpTION, the FE which corresponds to the exemplar frame LU is marked with *_Food* label, for instance the FE of **Ingredients**, *Product* is *Product_Food* in our annotation schema.

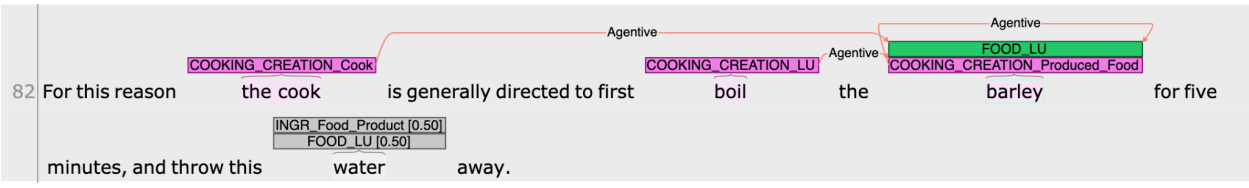
COOKING_CREATION: the agentive frame

Cooking_Creation Frame describes food and meal preparation. It is composed by a Cook creating a Produced_food from (raw) Ingredients. The Heating_instrument and/or the Container may also be specified. It has 12 frame elements: Cook, and Produced_food (Core), Container, Degree, Heating_Instrument, Ingredients, Manner, Means, Place, Purpose, Recipient, and Time (non-core). For the purpose of this study, we consider only the core FEs.

Core:

Cook [COOKING_CREATION_Cook]: The person who prepares the Produced_Food.

Produced_Food [COOKING_CREATION_Produced_Food]: The Produced_Food is the result of a Cook's efforts.



In this case the LU Food is represented by the FE Produced_Food through the *agentive* relation, while the Cooking Creation LUs are all those verbs which evoke the cooking process. A non-exhaustive list is provided below:

bake.v, baking.n, concoct.v, cook up.v, cook.n, cook.v, cooking.n, fix.v, fry.v, frying.n, grill.v, grilling.n, make.v, preparation.n, prepare.v, put together.v, roast.v, roasting.n, whip up.v

USING: the exemplar telic frame

This last frame is different from the others because it defines a prototypical frame for several other frames which present the same telic structure and represent the possible uses/purposes of food. I first present the **Using** frame and then I proceed with the other frames which share the same core structure.

For what concerns **Using**, we are again interested in its core construction thus we do not consider the non-core FEs of this frame. The core FEs of this Frame are:

(1) *Agent*: The Agent uses the Instrument to achieve their Purpose.

(2) *Instrument*: An entity directed by the Agent that serves to bring about the Purpose of the Agent.

(3) *Purpose*: This FE identifies the purpose for which an Agent uses the Instrument.

This frame has a list of lexical units which we provide below:

application.n, apply.v, employ.v, employment.n, exercise.v, operate.v, operation.n, operational.a, use.n, use.v, utilisation.n, utilization.n, utilise.v, utilize.v

We considered this frame because sometimes the purpose of food as for instance medicine or ritual object is not explicitly expressed by one of the LUs which identify the specific frames such as **Cure** or **Offering/Rite**. For instance, in the sentence 'ginger is sometimes used for sore throats' the **Cure** frame is not explicitly activated but we can

infer from the word 'sore throats' that ginger is used to sooth them, thus 'use' is considered a prototypical/superior LU for the **Cure** frame. We consider that the LUs for **Using** are part of the LUs of each of the frames we are considering in the telic relation if other elements of the sentence clearly identify the purpose among the frames we have individuated as alternative usages. This means that to the list of LUs of each frame in the telic relation construction we also add all the **Using** LUs. We proceed to present each of the frames, their LUs, and their FEs.

CURE: the medical telic frame

This frame deals with a *Healer* treating and curing an *Affliction* (the injuries, disease, condition or pain) of the *Patient*, when it mentions the use of a food item as a particular *Treatment* or *Medication*. This frame has 13 FEs: 6 core FEs and 7 non-core. For the purpose of the present study we consider the core FEs for this frame, aggregating together Medication and Treatment as food is nearer to the concept of treatment than medication, but do not exclude it. The final FEs are:

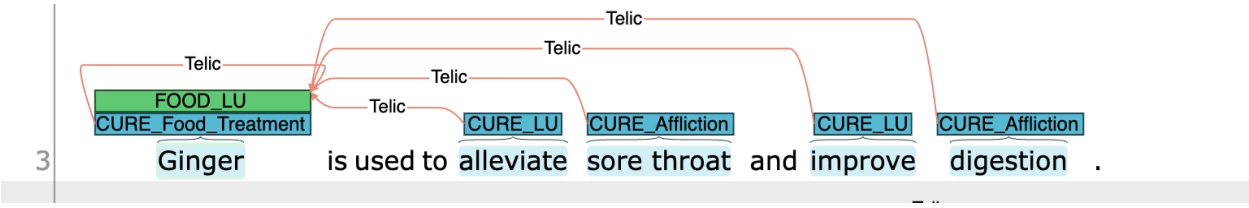
(1) *Affliction* [CURE_Affliction]: The Frame Element Affliction is generally the NP Object of a verb, frequently incorporating the Patient as a possessor.

(2) *Body_part* [CURE_Body_Part]: The Body_part is the specific area of the Patient's body which is treated.

(3) *Healer* [CURE_Healer]: The Healer represents anyone who treats or cures the Patient.

(4) *Treatment/Medication* [CURE_Food_Treatment]: The ingested, applied, injected, etc. substance designed to cure the Patient.

(5) *Patient* [CURE_Patient]: This is the sufferer of the injury, disease or pain and can occur as an NP Object in this frame. However, Patient may not be expressed as a separate Frame Element but frequently occurs as the possessor of the Affliction, an example of FE Conflation. The Patient may therefore not be tagged in a sentence although explicitly identified.



A list of LUs is provided below:

alleviate.v, alleviation.n, curable.a, curative.a, curative.n, cure.n, cure.v, ease.v, heal.v, healer.n, incurable.a, improve.v, nurse.v, palliate.v, palliation.n, palliative.a, palliative.n, rehabilitate.v, rehabilitation.n, rehabilitative.a, remedy.n, remedy.v, resuscitate.v, soothe.v, therapeutic.a, therapist.n, therapy.n, treat.v, treatment.n

As you can see from the example, LUs for Using have to be annotated just when no other frames are explicitly activated, otherwise the more specific LU has to be preferred.

INGESTION: the nutrition telic frame

In this frame, the purpose of food is that of being ingested. This frame formalizes that with an *Ingestor* consuming food or drink (*Ingestibles*), which entails putting the *Ingestibles* in the mouth for delivery to the digestive system.

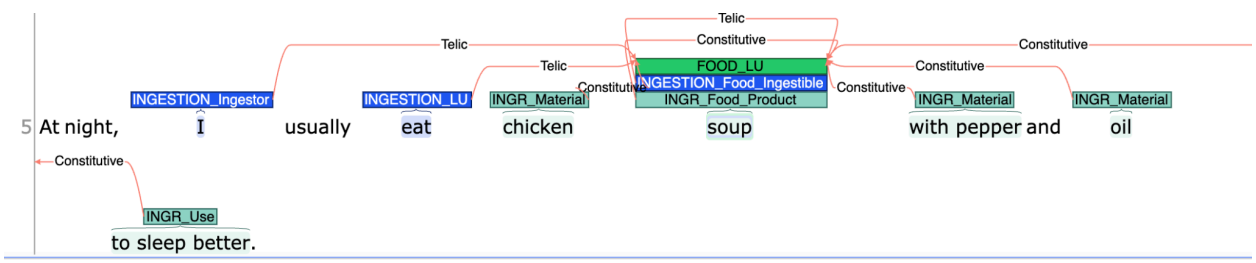
The core FEs of this frame that we considered in this study are:

(1) *Ingestibles* [INGESTION_Food_Ingestible]: The Ingestibles are the entities that are being consumed by the Ingestor.

(2) *Ingestor* [INGESTION_Ingestor]: The Ingestor is the person eating or drinking.

The LUs are all those verbs or noun which refer to the act of eating and drinking

breakfast.v, consume.v, devour.v, dine.v, down.v, drink.v, eat.v, feast.v, feed.v, gobble.v, gulp.n, gulp.v, guzzle.v, have.v, imbibe.v, ingest.v, ingestion.n, lap.v, lunch.v, munch.v, nibble.v, nosh.v, nurse.v, put away.v, put back.v, quaff.v, sip.n, sip.v, slurp.n, slurp.v, snack.v, sup.v, swig.n, swig.v, swill.v, tuck.v



OFFERING/RITE: the ritualistic telic frame

In this case to express the possibility of food being offered as part of a rite or as a luxury gift in a specific historical period we rely on two frames, aggregating their LUs and core FEs.

In the **Offering Frame**, an *Offerer* indicates that he or she is able and willing to give a *Theme* to a *Potential_recipient*. In the absence of further qualifications, it is often understood that the *Potential_recipient* accepts the *Theme*.

The **Rite** Frame concerns instead rituals performed in line with religious beliefs or tradition.

To the **<u>Offering</u>** Core FEs we also include some aspects of **Rite**:

(1) *<u>Offerer</u>* [OFFER_RITE_Offerer]: The person that begins in possession of the Theme and offers it to the Potential_recipient.

(2) *Potential recipient* [OFFER_RITE_Recipient]: The entity that ends up in possession of the Theme.

(3) *Theme/Object/Instrument* [OFFER_RITE_Food_Theme]: The object which is offered for transfer, used to perform a particular rite, or on which a rite is performed.

(4) *Type* [OFFER_RITE_Type]: The type of ritual or ceremony being performed.

The list of LUs of both the frames harmonized according to our purpose are provided below, with the addition of gift.n and gift.v from the more general frame **Giving**

extend.v, offer.v, proffer.v, serve.v, gift.n, gift.v, anoint.v, baptism.n, baptize.v, bar mitzvah.n, bless.v, blessing.n, christen.v, christening.n, circumcise.v, circumcision.n, communion.n, confession.n, confirm.v, confirmation.n, consecrate.v, consecration.n, eucharist.n, evensong.n, exercise.n, initiate.v, initiation.n, mass.n, ordain.v, order.v, ordination.n, pray.v, prayer.n, rite of passage.n, rite.n, ritual.n, ritualistic.a, sacrament.n, sacrifice.n, sacrifice.v, service.n, unction.n, vesper.n, vigil.n, worship.n

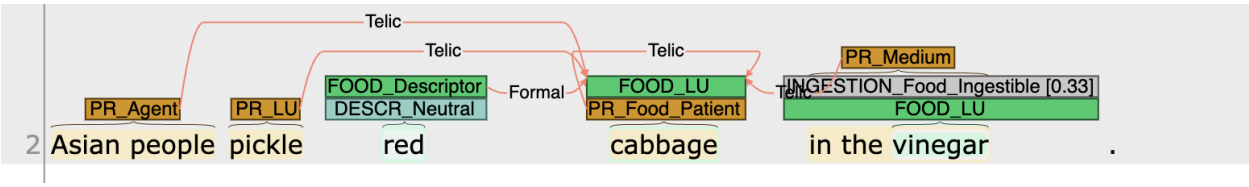
PRESERVING: the preservation telic frame

In this frame an *Agent* preserves a *Patient* (food) in order to prevent it from decaying. A *Medium* may be used.

(1) *Agent* [PR_Agent]: The Agent is the person performing the intentional act that leads to the preservation.

(2) *Medium* [PR_Food_Medium]: Medium is the substance in which the Patient might be submerged or treated with to be preserved.

(3) *Patient* [PR_Patient]: The Patient is organic or non-organic matter that undergoes preservation.



Possible Labels

Relation: *Formal*

Frame: Food

- FOOD_Food_LU: Words referring to food items (i.e., LUs in the Food frame).
- FOOD_Constituent_Part: A component or a part of the FOOD_LU.
- FOOD_Descriptor: A characteristic or classification of the FOOD_LU, including both descriptive and typological information. It is further annotated for intrinsic polarity: (1) Positive [DESCR_Positive]; (2) Negative [DESCR_Negative]; (3) Neutral [DESCR_Neutral].
- Count FOOD_Count: The number of units in which the FOOD_LU is quantified.
- Unit FOOD_Unit: The standard unit of measurement used to quantify the FOOD_LU (e.g., grams, liters).

Relation: *Constitutive*

Frame: Ingredients

- INGR_Material_LU: The entity which is used to make a Product.
- INGR_Food_Product: The entity or substance that is created out of the Material, in this case the food.
- INGR_Use: The purpose for which the Material is intended.

Relation: *Agentive*

Frame: Cooking_Creation

- COOKING_CREATION_LU Every word which expresses the action of cooking.
- COOKING_CREATION_Cook The person who prepares the Produced_Food.
- COOKING_CREATION_Produced_Food It is the result of a Cook's efforts.

Relation: *Telic*

Frame: Using_Cure

- CURE_LU: Every word expressing the act of treating or curing someone.
- CURE_Affliction: It represents the injury, disease, condition, or pain felt by the patient.
- CURE_Healer: It represents anyone who treats or cures the Patient.
- CURE_Food_Treatment The ingested, applied, injected, etc. substance designed to cure the Patient.
- CURE_Patient It is the sufferer of the injury, condition, disease or pain.

Frame: Using_Preserving

- PR_LU Every word expressing the act of preserving something from decaying.
- PR_Agent The person performing the intentional act that leads to the preservation.
- PR_Food_Medium The substance in which the Patient might be submerged or treated with to be preserved.
- PR_Patient The organic or non-organic matter that undergoes preservation.

Frame: Using_Offering/Rite

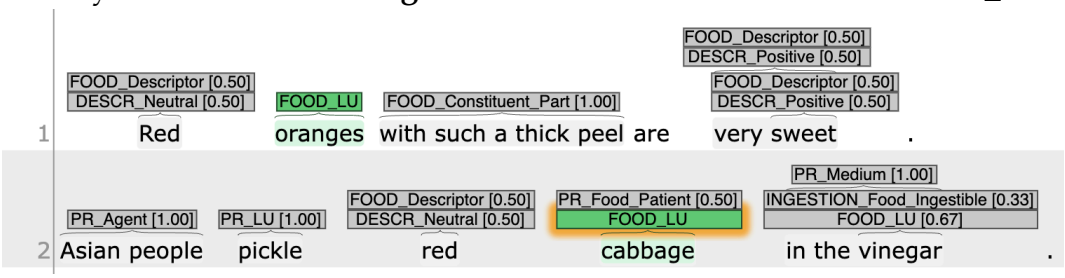
- OFFER_RITE_LU
- OFFER_RITE_Offerer The person that begins in possession of the Theme and offers it to the Potential_recipient.
- OFFER_RITE_Recipient The entity that ends up in possession of the Theme.
- OFFER_RITE_Food_Theme The object which is offered for transfer, used to perform a particular rite, or on which a rite is performed.
- OFFER_RITE_Type The type of ritual or ceremony being performed.

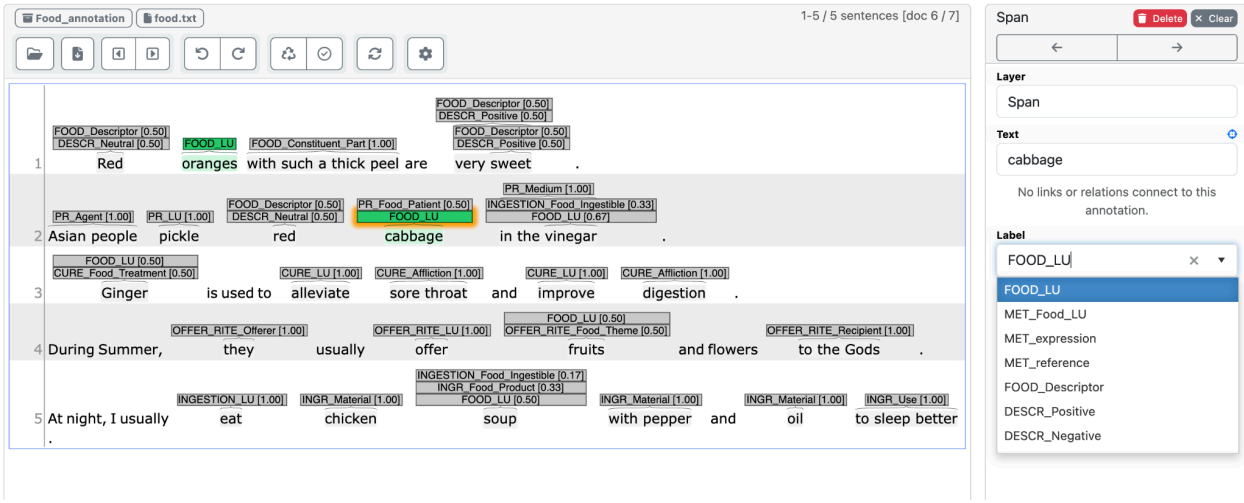
Frame: Using_Ingestion

- INGESTION_LU Every word expressing the act of eating or drinking.
- INGESTION_Food_Ingestible The Ingestibles are the entities that are being consumed by the Ingestor.
- INGESTION_Ingestor The Ingestor is the person eating or drinking

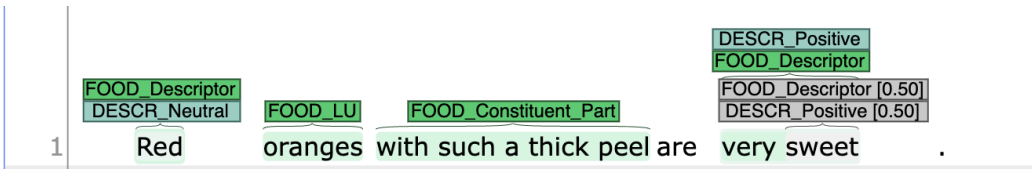
Annotation Workflow

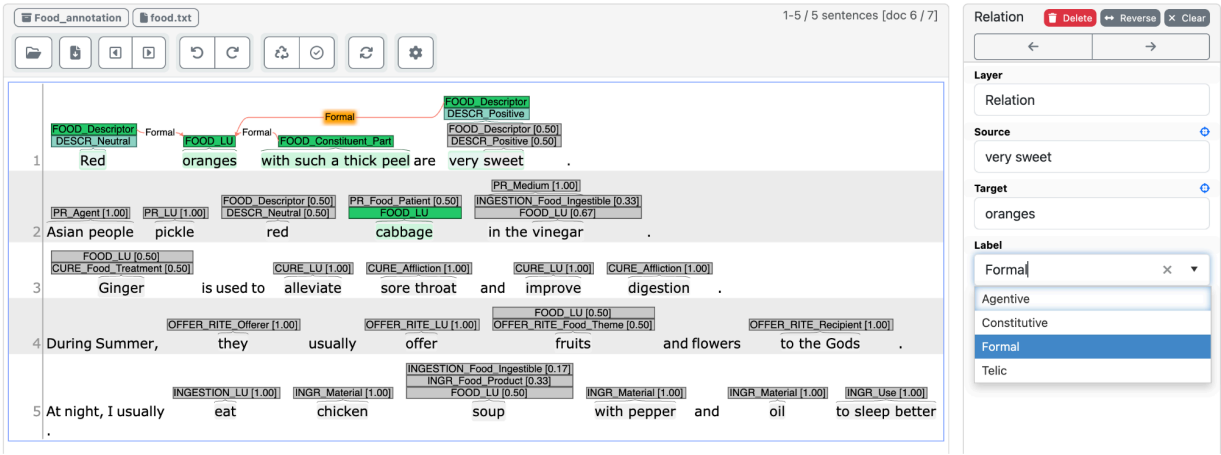
1) In a given sentence, the annotator has to:
    - Identify the **words referring to food** and annotate them as FOOD_LU.





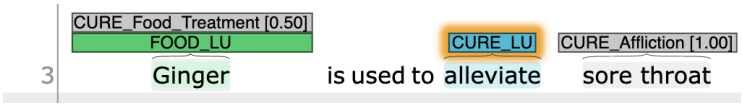
- Annotate **Frame Elements (FEs)** related to each FOOD_LU in the **FOOD frame**, and link them to the LU using the relation *formal*.



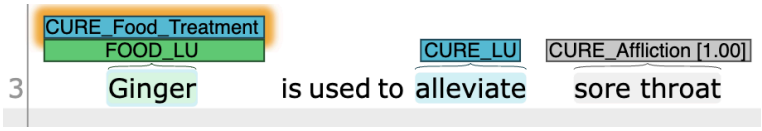


2) Check whether the food item **participates in another event or usage** that activates another frame (e.g., CURE, INGESTION, PRESERVING), namely if any other LUs or food realizations is present . If so:

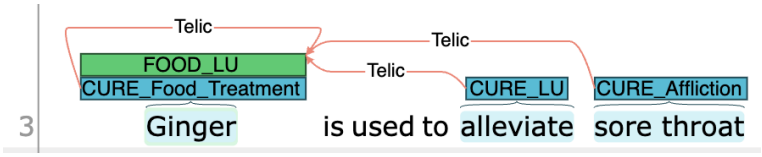
- Annotate the predicate (e.g., *alleviates*) or any other argument of other frames (e.g., *chicken* as Ingredient) as the LU of that frame, e.g., 'alleviates' as CURE_LU.



- Re-annotate the food item with its new role in the frame (e.g., CURE_Food_Treatment for *ginger*).



- Link the frame's participants to the corresponding FOOD_LU via the relation (here *telic*). Note that also the FE-to-LU relation (double span annotation) has to be annotated.



Frame	FE/LU	Label	Definition
Food	**Food** (LU)	FOOD_Food_LU	Words referring to food items (i.e., LUs in the Food frame).
	Constituent_parts	FOOD_Constituent_Part	A component or a part of the FOOD_LU.
	Descriptor	FOOD_Descriptor	A characteristic or classification of the FOOD_LU, including both descriptive and typological information. It is further annotated for intrinsic polarity: (1) Positive [DESCR_Positive]; (2) Negative [DESCR_Negative]; (3) Neutral [DESCR_Neutral].
	Count	FOOD_Count	The number of units in which the FOOD_LU is quantified.
	Unit	FOOD_Unit	The standard unit of measurement used to quantify the FOOD_LU (e.g., grams, liters).
Ingredients	**Material** (LU)	INGR_Material_LU	The entity which is used to make a Product.
	Product	INGR_Food_Product	The entity or substance that is created out of the Material, in this case the food.
	Use	INGR_Use	The purpose for which the Material is intended.
Cooking_Creation	LU	COOKING_CREATION_LU	Every word which expresses the action of cooking.
	Cook	COOKING_CREATION_Cook	The person who prepares the Produced_Food.
	Produced_Food	COOKING_CREATION_Produced_Food	It is the result of a Cook's efforts.
Using_Cure	LU	CURE_LU	Every word expressing the act of treating or curing someone
	Affliction	CURE_Affliction	It represent the injury, disease, condition, or pain felt by the patient
	Healer	CURE_Healer	It represents anyone who treats or cures the Patient.
	Treatment/ Medication	CURE_Food_Treatment	The ingested, applied, injected, etc. substance designed to cure the Patient.
	Patient	CURE_Patient	It is the sufferer of the injury, condition, disease or pain.
Using_Preserving	LU	PR_LU	Every word expressing the act of preserving something from decaying.
	Agent	PR_Agent	The person performing the intentional act that leads to the preservation.
	Medium	PR_Food_Medium	The substance in which the Patient might be submerged or treated with to be preserved.
	Patient	PR_Patient	The organic or non-organic matter that undergoes preservation.
Using_Offering/ Rite	LU	OFFER_RITE_LU	Every word that expresses the act of offering, gifting, or performing a ritualistic action involving food.
	Offerer	OFFER_RITE_Offerer	The person that begins in possession of the Theme and offers it to the Potential_recipient.
	Potential_recipient	OFFER_RITE_Recipient	The entity that ends up in possession of the Theme.
	Theme/Object/ Instrument	OFFER_RITE_Food_Theme	The object which is offered for transfer, used to perform a particular rite, or on which a rite is performed.
	Type	OFFER_RITE_Type	The type of ritual or ceremony being performed.
Using_Ingestion	LU	INGESTION_LU	Every word expressing the act of eating or drinking.
	Ingestibles	INGESTION_Food_Ingestible	The Ingestibles are the entities that are being consumed by the Ingestor.
	Ingestor	INGESTION_Ingestor	The Ingestor is the person eating or drinking.

Table 1: Frames for Contextual Food Domain, FEs and definitions. In the Offering/Rite, we underlined the FEs taken from FrameNet Offering, while the rest are taken from FrameNet Rite. In bold, we underline the core FEs of each frame.